

TIC 446502638

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.107957 d	VSX	no match
Epoch (t0)	3480.6522 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	0.231 % (2,310 ppm)	Gaia variable classification	no match
SNR	12.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	16 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	0.91 (gate >= 0.50)		
Robust depth sigma	9.9 (gate >= 7.0)		
Odd/even z-score	0.18 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

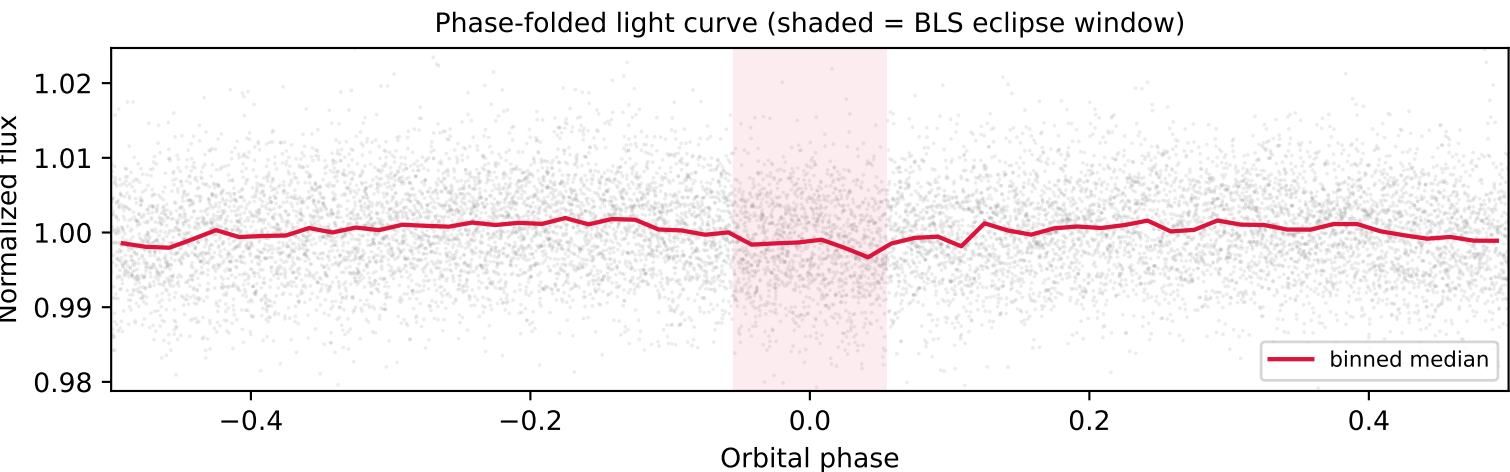
Sector 80 light-curve dip in this sector: 11.1 sigma t0 refined by 0.0300 phase

Difference-image peak SNR 6.06 (gate >= 5.0) Compactness 0.56 (gate >= 0.50)

Dimming centroid offset 0.16 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1748977177

0.25 px away, dTmag +6.56 - farther from the dimming than the target



difference-image figure not found

Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.090 px (dimming centroid 0.158 px from target vs 0.247 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.